

CS 546

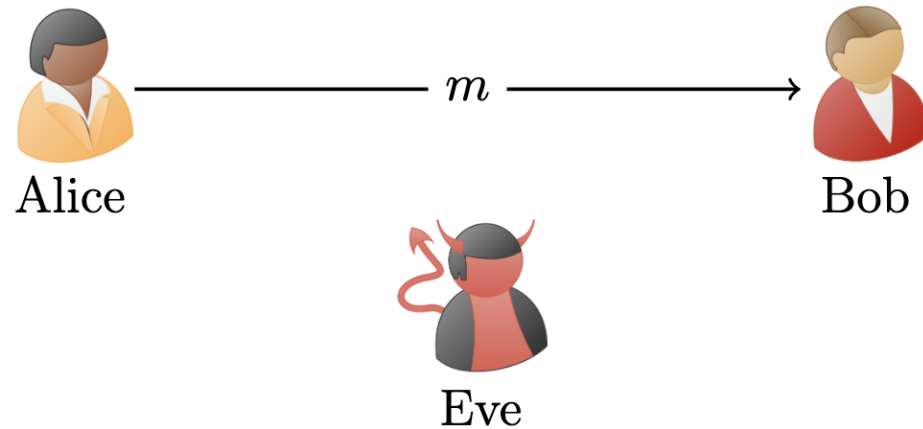
Computer System Security

Lecture 1.3: Crypto (Confidentiality - I)

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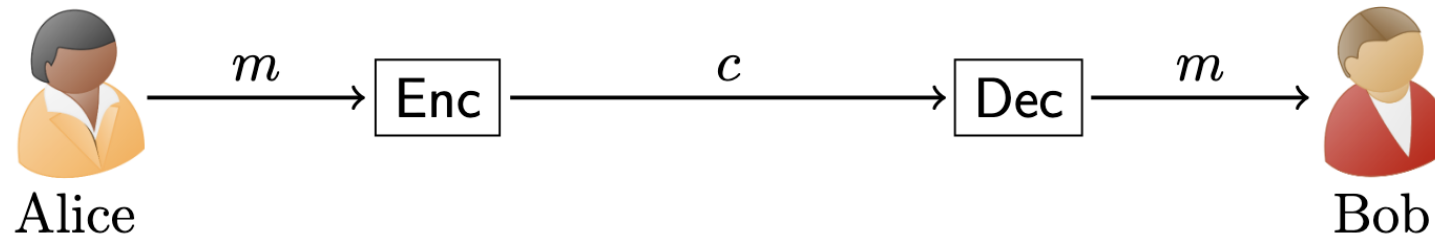
Fall 2026

Problem Setting: Secret Communication



- Alice wants to send a message m to Bob.
- How can Alice convey this message to Bob, while keeping it hidden from an eavesdropper Eve?
- In other words, how can we ensure confidentiality of m ?

Problem Setting: Secret Communication



- **Encryption:** Alice transforms (encrypts) the message m – also called the “plaintext” – into a “ciphertext” c .
- **Communication:** Alice sends c to Bob.
- **Decryption:** Bob recovers (decrypts) the original m from c .

Kerckhoffs' Principle

- **Goal:** Bob should be able to decrypt c , but not Eve.
- To enable that, something must be kept secret from Eve.
- Should we keep the details of the Enc and Dec algorithms secret from Eve?

No! Why not?

- If Eve eventually learns the details of Enc and Dec, we will have to invent new algorithms.
- Inventing good encryption algorithms is not an easy task.

Kerckhoffs' Principle

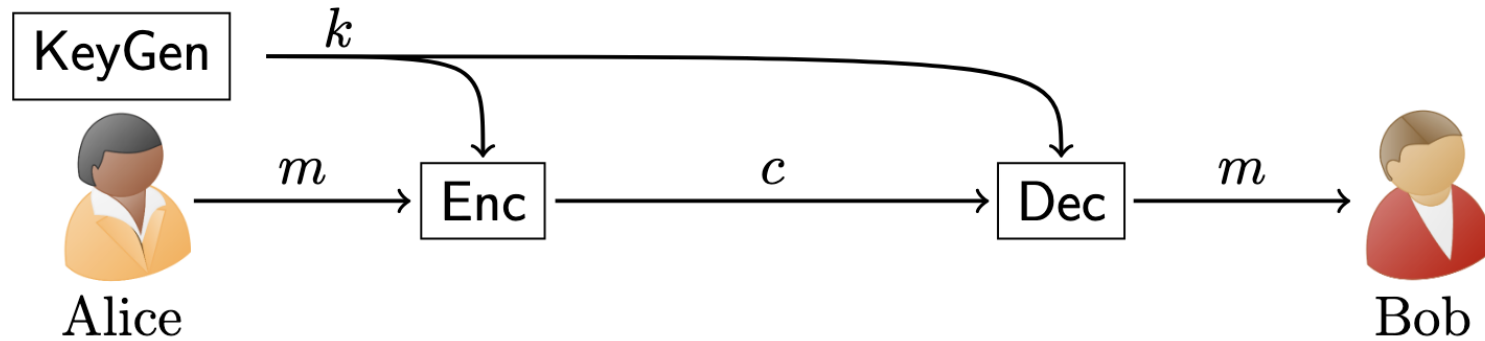
“The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience.”

- If the algorithms themselves are not secret, then there must be some other secret information in the system.
- That information is called the **secret key**.
- If a secret key gets compromised, we only need to choose a new one, not reinvent a new encryption algorithm.
- Put differently, the security of the system is concentrated in the secrecy of the key, not the algorithm.



Auguste Kerckhoffs
19th-century
Dutch cryptographer

Syntax of Encryption Schemes



- **Key Generation:** KeyGen is the randomized process of choosing a secret key k , which is used as input to both Enc and Dec algorithms.

Kerckhoffs was right because ...

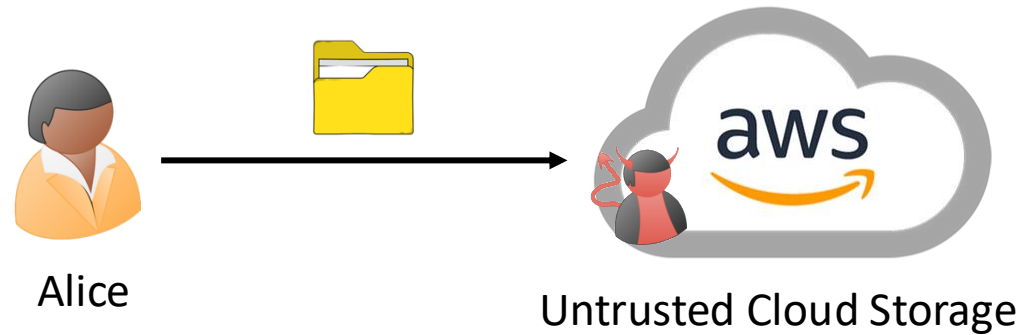
- Maintaining privacy of a short key (≈ 100 bits) is easier than maintaining privacy of a large algorithm.
- Easy to replace the key than a whole program if exposed.
- It is infeasible to imagine a secret encryption scheme for every pair of communicating parties.
- Open designs undergo public scrutiny, and hence, are likely to be stronger.
- Security flaws (if they exist) can be revealed by ethical hackers.
- Public designs enable establishment of standards.



**Dangerous to use proprietary
Encryption schemes!**

Another Application of Encryption

(Secure Storage)



- Alice wants to store some private data on the cloud, but she does not trust the cloud service provider.
- Alice can generate an encryption key, encrypt her data using this key and send encrypted data to the cloud. She keeps the key with herself.

Classical Ciphers/Encryption Mechanisms

(Shift Cipher (~58 BC))

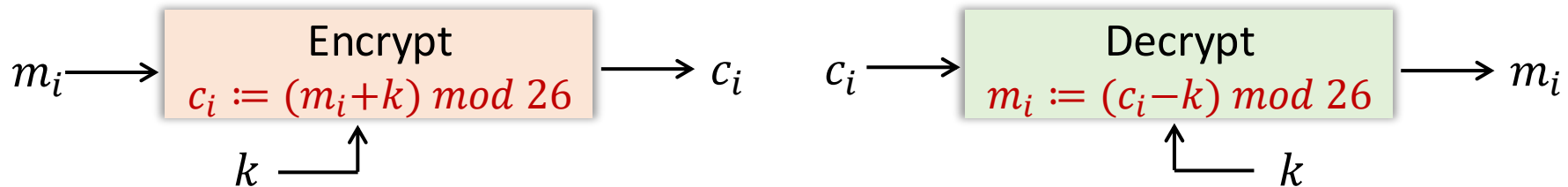


KeyGen: Choose a random key $k \in \{0, 1, 2, \dots, 25\}$.

Encryption Algorithm: Shift by k .

Example: $k = 3$

msg	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
cipher	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C



Broken

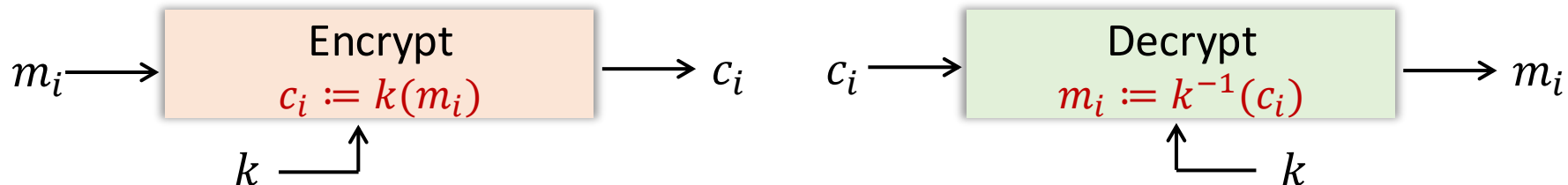
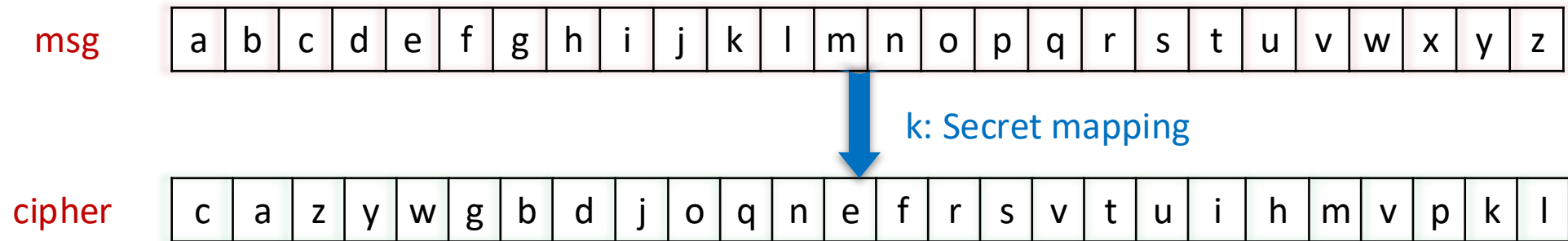
Brute-force attack! Try all 26 possible keys

Classical Ciphers/Encryption Mechanisms

(Mono-Alphabetic Substitution Cipher (1300s))

KeyGen: Let $\mathcal{K}$ be all permutations on the set $\{0,1,2, \dots, 25\}$. Choose a random key $k \in \mathcal{K}$.

Encryption Algorithm: Shift using permutation k .



$|\mathcal{K}| = 26! = 2^{88}$. Hence a brute-force attack is infeasible!

Classical Ciphers/Encryption Mechanisms

(Mono-Alphabetic Substitution Cipher (1300s))

Still Broken

Frequency/Statistical Analysis

We can exploit statistical patterns in the english language to break this cipher.

UKBYBIPOUZBCUFEEBORUKBYBHOBBERFESPVKBWFOFERVNBCVBZPRUBOFERVNBCVBPCYYFVUFO
FEIKNWFRFIKJNUPWRFIPOUNVNIPUBRNCUKBEFWWFDNCHXCBOHOPYXPUBNCUBOYNRVNIWN
CPOJIOFHOPZRVFZIXUBORJRUBZRBCHNCBBONCHRJZSFWNVRJRUBZRPCYZPUKBZPUNVPWPCYVF
ZIXUPUNFCPWVRVNBCVBRPYYNUNFCPWWJUKBYBIPOUZBCUIPOUNVNIPUBRNCCHOPYXPUBNCUB
OYNRVNIWNCPOJIOFHOPZRNCRVNBCUNENVVFZIXUNCHPCYVFZIXUPUNFCPWZPUKBZPUNVR

B	36	→ E
N	34	
U	33	→ T
P	32	→ A
C	26	

NC	11	→ IN
PU	10	→ AT
UB	10	
UN	9	

digrams

UKB	6	→ THE
RVN	6	
FZI	4	

trigrams

Classical Ciphers/Encryption Mechanisms

(Vigenère: Poly-alphabetic Shift Cipher (1300s))

KeyGen: Key k is a random word of length t .

Encryption Algorithm: as an example, say $k = CRYPTO$

msg

w	h	a	t	a	n	i	c	e	d	a	y	t	o	d	a	y
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

+ mod 26

key

C	R	Y	P	T	O	C	R	Y	P	T	O	C	R	Y	P	T
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

cipher

Z	Z	Z	J	U	C	L	U	D	T	U	N	W	G	C	Q	S
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



Exercise

Think of an attack!

This cipher was a lot harder to break. In fact, a systematic attack took years to devise.



"I got your email. Was it encrypted or is
your spelling *that* bad?"